PREPPULSE AI

Next-Generation Tech Placement & Mindset Hub

- Domain: Artificial Intelligence & Web Engineering
- Key Innovations: Real-time Voice Mock Interviews with Gemini AI
- Comprehensive Practice: DSA, Core CS (OS, DBMS, CN), System Design
- Cloud Architecture: Firebase Firestore Persistent Sync & Role Security
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- Academic Year: 2025–2026 | B.Tech Computer Science & Engineering

Problem Statement & Motivation

Bridging the Gap Between Academics and Tech Hiring

- Tech interviews require dual competence: conceptual depth and verbal communication.
- Traditional placement preparation is fragmented across multiple disparate websites.
- Mock interviews with human mentors are expensive and lack scalable on-demand availability.
- Candidates suffer from interview anxiety without realistic voice-based mock feedback.
- Solution: An integrated AI platform providing adaptive voice coaching and progress analytics.

Project Objectives & Scope

Target Capabilities & Deliverables

- 1. Intelligent Voice Mock Interviewer: Real-time audio capture, transcription, and scoring.
- 2. Core CS Conceptual Bank: Curated question sets for DBMS, OS, Computer Networks, and OOP.
- 3. Interactive DSA Sandbox: Problem categorizations, time/space complexity notes, and hints.
- 4. LeetCode Profile Synchronization: Live problem stats, difficulty breakdown, and daily streaks.
- 5. Secure Cloud Synchronization: Cloud Firestore persistence with strict role security rules.

System Architecture & Tech Stack

Modular Client-Server & Cloud Infrastructure

- Frontend: React 19, TypeScript, Tailwind CSS v4, Motion animation primitives.
- Backend / API Layer: Express.js, TypeScript Execution (tsx), Vite integration.
- AI / ML Engine: Google Gemini API with smart model cascade & automatic error recovery.
- Database & Auth: Firebase Authentication (Google/Email) & Cloud Firestore.
- Audio Processing: Web Audio API, MediaRecorder, PCM transcription pipeline.

Voice Mock Interview Module

Speech-to-Text & Gemini AI Evaluation Engine

- Dynamic question generation tailored to company profiles (FAANG, Unicorns, Consultancies).
- Dual input modes: Speech recording via MediaRecorder with text input fallback.
- Audio Transcription: High-fidelity voice conversion with adaptive timeout guards.
- Multi-dimensional rubric scoring: Technical accuracy, communication clarity, and completeness.
- Actionable feedback breakdown: Key strengths, areas for improvement, and ideal answers.

Core CS & DSA Practice Engine

Comprehensive Computer Science Knowledge Hub

- Subject Modules: Operating Systems (Threads, Virtual Memory, Deadlocks).
- Database Management: Normalization, Indexing (B+ Trees), ACID, Transaction concurrency.
- Computer Networks: OSI/TCP-IP, DNS resolution, Socket programming, Protocols.
- Object-Oriented Programming: Polymorphism, Design Patterns (SOLID), Encapsulation.
- Algorithmic Problem Bank: Curated DSA sheets with complexity analysis and edge cases.

LeetCode Sync & Executive Analytics

Performance Metrics & Readiness Visualizer

- Live LeetCode profile scraper and statistics aggregation (Easy / Medium / Hard).
- Executive Dashboard: Interview Readiness Index calculation and milestone tracking.
- Radar charts and category progress indicators highlighting weak topic areas.
- Streak calculation and daily goal tracker fostering disciplined preparation habits.
- Historical interview log: Track performance trajectory across practice sessions.

Database Design & Security Rules

Zero-Leak Abstraction & Role-Based Access Control

- Configuration Abstraction Layer (firebase.config.ts) prevents credential leaks on GitHub.
- Cloud Firestore Schema: Strict subcollection hierarchy for user mock interviews and progress.
- Security Rules (firestore.rules): Enforced user ownership (request.auth.uid == userId).
- Sanitized templates (.env.example & firebase-applet-config.example.json) for open source.
- Zero-client secret exposure: Sensitive Gemini API calls handled securely via backend proxy.

Testing, Benchmarking & Results

Rigorous Validation of System Performance

- AI Evaluation Consistency: Validated across 100+ standard engineering interview questions.
- Speech Recognition Accuracy: ~96% transcription fidelity on technical programming terms.
- Sub-2s Response Latency: Parallelized prompt engineering and intelligent model fallbacks.
- Zero-Fatal Error Resilience: Graceful fallback to text mode when microphone is unavailable.
- Cross-Browser Compatibility: Verified on Chrome, Edge, Safari, and mobile viewports.

Conclusion & Future Scope

Summary of Achievements & Next Steps

- Successfully developed an enterprise-grade Tech Placement & Interview Preparation Hub.
- Empowers students with personalized, AI-driven, unbiased voice mock interview practice.
- Unified DSA, Core CS, and coding metrics into a cohesive, responsive web platform.
- Future Enhancements: Real-time video body language/eye contact analysis via Computer Vision.
- Future Enhancements: Peer-to-peer live mock matching and automated resume ATS scanner.